

Si5351A Clock Configuration

Tonnere Board — HDMI Video Modes

Reference clock: 25 MHz XTAL. Library: MR-DOS/Si5351-lib (STM32). All frequencies computed from AN619 equations. Two VCO configurations cover all HDMI output modes. Three Si5351A outputs are used per device: CLK0 = pixel clock, CLK1 = Atari system clock (7 MHz), CLK2 = colour subcarrier (fsc).

Signal Chain

```
XTAL 25 MHz
|
+--[ PLLA: x(a+b/c) ]---[ MS0 / CLK0 ]--> Pixel clock (27.000 or 74.250 MHz)
|
+--[ MS1 / CLK1 ]--> Atari system clock (~7.09 / ~7.16 MHz)
|
+--[ MS2 / CLK2 ]--> Colour subcarrier fsc (~4.43 / ~3.58 MHz)
```

PLL A — VCO Configurations

Output modes	VCO	PLL a	b	c	P1	P2	P3
576p/i PAL@50, 480p/i NTSC@59.94	810.0 MHz	32	2	5	3635	1	5
720p/1080i PAL@50, NTSC@60	742.5 MHz	29	7	10	3289	6	10

Output Frequency Summary

Mode	Pixel clk	Atari Hz	Atari MHz	Err (ppm)	fsc Hz	fsc MHz	Err (ppm)	fsc MS a+b/c
576p/i PAL@50	27.000M	7,093,788.8	7.0937888	+0.1156	4,433,618.87	4.4336189	+0.0277	182+139/200
480p/i NTSC@59.94	27.000M	7,159,090.9	7.1590909	+0.1270	3,579,545.45	3.5795455	+0.1270	226+2/7
720p/1080i PAL@50	74.250M	7,093,786.5	7.0937865	-0.2094	4,433,618.72	4.4336187	-0.0073	167+167/355
720p/1080i PAL@50 Hi	74.250M	7,093,788.0	7.0937880	-0.0004	4,433,618.72	4.4336187	-0.0073	167+167/355
720p/1080i NTSC@60	74.250M	7,159,090.9	7.1590909	+0.1270	3,579,545.45	3.5795455	+0.1270	207+3/7
720p/1080i NTSC@60 Hi	74.250M	7,159,089.9	7.1590899	-0.0109	3,579,545.45	3.5795455	+0.1270	207+3/7

Hi rows use large-denominator fractional MS for sub-0.01 ppm accuracy. Standard rows use small denominators (lower jitter, <0.21 ppm). 'Hi' suffix = HiAcc variant.

AN619 Register Parameters (P1 / P2 / P3)

Raw values to write directly to Si5351 registers, computed per AN619: P1 = 128a + floor(128b/c) - 512, P2 = 128b - c*floor(128b/c), P3 = c.

Mode	Output	a	b	c	P1	P2	P3
576p/i PAL@50	MS0 pixel	30	0	1	3328	0	1
	MS1 Atari	114	26	141	14103	85	141
	MS2 fsc	182	139	200	22872	192	200
480p/i NTSC@59.94	MS0 pixel	30	0	1	3328	0	1
	MS1 Atari	113	1	7	13970	2	7
	MS2 fsc	226	2	7	28452	4	7

720p/1080i PAL@50	MS0 pixel	10	0	1	768	0	1
	MS1 Atari	104	93	139	12885	89	139
	MS2 fsc	167	167	355	20924	76	355
720p/1080i PAL@50 Hi	MS0 pixel	10	0	1	768	0	1
	MS1 Atari	104	6382	9539	12885	6081	9539
	MS2 fsc	167	167	355	20924	76	355
720p/1080i NTSC@60	MS0 pixel	10	0	1	768	0	1
	MS1 Atari	103	5	7	12763	3	7
	MS2 fsc	207	3	7	26038	6	7
720p/1080i NTSC@60 Hi	MS0 pixel	10	0	1	768	0	1
	MS1 Atari	103	7138	9993	12763	4301	9993
	MS2 fsc	207	3	7	26038	6	7

MR-DOS/Si5351-lib — Struct Configuration

PLLA (VCO)

```
// 810 MHz VCO (576p PAL@50, 480p NTSC@59.94)
Si5351_Config.PLL[PLL_A].PLL_Multiplier_Integer = 32;
Si5351_Config.PLL[PLL_A].PLL_Multiplier_Numerator = 2;
Si5351_Config.PLL[PLL_A].PLL_Multiplier_Denominator = 5;
Si5351_Config.PLL[PLL_A].PLL_Clock_Source = PLL_Clock_Source_XTAL;

// 742.5 MHz VCO (720p / 1080i)
Si5351_Config.PLL[PLL_A].PLL_Multiplier_Integer = 29;
Si5351_Config.PLL[PLL_A].PLL_Multiplier_Numerator = 7;
Si5351_Config.PLL[PLL_A].PLL_Multiplier_Denominator = 10;
Si5351_Config.PLL[PLL_A].PLL_Clock_Source = PLL_Clock_Source_XTAL;
```

MS0 / CLK0 — Pixel Clock

```
// MS0 -> CLK0: pixel clock (always integer divide)
Si5351_Config.MS[MS0].MS_Clock_Source = MS_Clock_Source_PLLA;
Si5351_Config.MS[MS0].MS_Divider_Integer = 30; // 27.000 MHz (810 VCO)
// or = 10; // 74.250 MHz (742.5 VCO)
Si5351_Config.MS[MS0].MS_Divider_Numerator = 0;
Si5351_Config.MS[MS0].MS_Divider_Denominator = 1;

Si5351_Config.CLK[CLK0].CLK_Clock_Source = CLK_Clock_Source_MS_Own;
Si5351_Config.CLK[CLK0].CLK_Enable = ON;
Si5351_Config.CLK[CLK0].CLK_PowerDown = OFF;
Si5351_Config.CLK[CLK0].CLK_I_Drv = CLK_I_Drv_8mA;
```

MS1 / CLK1 — Atari System Clock

```
Si5351_Config.MS[MS1].MS_Clock_Source = MS_Clock_Source_PLLA;

// 576p/i PAL@50 114 + 26/141 (+0.116 ppm)
Si5351_Config.MS[MS1].MS_Divider_Integer=114; .Numerator=26; .Denominator=141;

// 480p/i NTSC@59.94 113 + 1/7 (+0.127 ppm)
Si5351_Config.MS[MS1].MS_Divider_Integer=113; .Numerator=1; .Denominator=7;

// 720p/1080i PAL@50 104 + 93/139 (-0.209 ppm)
Si5351_Config.MS[MS1].MS_Divider_Integer=104; .Numerator=93; .Denominator=139;

// 720p/1080i PAL@50 Hi 104 + 6382/9539 (-0.0004 ppm)
Si5351_Config.MS[MS1].MS_Divider_Integer=104; .Numerator=6382; .Denominator=9539;

// 720p/1080i NTSC@60 103 + 5/7 (+0.127 ppm)
```

```

Si5351_Config.MS[MS1].MS_Divider_Integer=103; .Numerator=5; .Denominator=7;

// 720p/1080i NTSC@60 Hi 103 + 7138/9993 (-0.011 ppm)
Si5351_Config.MS[MS1].MS_Divider_Integer=103; .Numerator=7138; .Denominator=9993;

Si5351_Config.CLK[CLK1].CLK_Clock_Source = CLK_Clock_Source_MS_Own;
Si5351_Config.CLK[CLK1].CLK_Enable      = ON;
Si5351_Config.CLK[CLK1].CLK_PowerDown  = OFF;
Si5351_Config.CLK[CLK1].CLK_I_Drv      = CLK_I_Drv_8mA;

```

MS2 / CLK2 — Colour Subcarrier (fsc)

```

// PAL fsc from 810 MHz VCO    182 + 139/200 (+0.028 ppm)
// P1=22872, P2=192, P3=200
Si5351_Config.MS[MS2].MS_Divider_Integer=182; .Numerator=139; .Denominator=200;

// NTSC fsc from 810 MHz VCO    226 + 2/7      (+0.127 ppm)
// P1=28452, P2=4, P3=7
Si5351_Config.MS[MS2].MS_Divider_Integer=226; .Numerator=2; .Denominator=7;

// PAL fsc from 742.5 MHz VCO    167 + 167/355 (-0.007 ppm)
// P1=20924, P2=76, P3=355
Si5351_Config.MS[MS2].MS_Divider_Integer=167; .Numerator=167; .Denominator=355;

// NTSC fsc from 742.5 MHz VCO    207 + 3/7      (+0.127 ppm)
// P1=26038, P2=6, P3=7
Si5351_Config.MS[MS2].MS_Divider_Integer=207; .Numerator=3; .Denominator=7;

Si5351_Config.MS[MS2].MS_Clock_Source    = MS_Clock_Source_PLLA;
Si5351_Config.CLK[CLK2].CLK_Clock_Source = CLK_Clock_Source_MS_Own;
Si5351_Config.CLK[CLK2].CLK_Enable      = ON;
Si5351_Config.CLK[CLK2].CLK_PowerDown  = OFF;
Si5351_Config.CLK[CLK2].CLK_I_Drv      = CLK_I_Drv_8mA;

```

Initialisation Call Sequence

```

Si5351_StructInit(&Si5351_Config);      // zero + defaults first
// ... fill all fields as above ...
Si5351_OSCConfig(&Si5351_Config);       // xtal load capacitance
Si5351_PLLConfig(&Si5351_Config, PLL_A);
Si5351_PLLReset(&Si5351_Config, PLL_A);
Si5351_MSConfig(&Si5351_Config, MS0);
Si5351_MSConfig(&Si5351_Config, MS1);
Si5351_MSConfig(&Si5351_Config, MS2);
Si5351_CLKConfig(&Si5351_Config, CLK0);
Si5351_CLKConfig(&Si5351_Config, CLK1);
Si5351_CLKConfig(&Si5351_Config, CLK2);

```

Note: Si5351_StructInit sets CLK_PowerDown=ON for all outputs by default. Explicitly set CLK_PowerDown=OFF for each output you are using before calling CLKConfig.